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# Preference Training

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30 October 2025



# Supervised Training



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- So far: supervised learning
  - given source sentence
  - predict target sentence, word by word
  - compare against each word in the reference translation
  - loss: negative log probability given to the reference word
- This is done for
  - training neural machine translation models
  - pre-training large language models
  - fine-tuning large language models

# Reinforcement Learning

- In some tasks, we know success only after many steps
- Example: chess
  - many moves
  - win (“reward”) only known at end
- Translation
  - are getting individual words right sufficient?
  - ... or does it matter that the whole sentence translation is good?



- Jargon

**Reward** Score for a prediction: could be binary: win/loss

**Policy** Underlying decision mechanism for steps — in our case this is the MT model or LLM

**Policy Gradient** Learning method that adjusts the model (policy) parameters directly in the direction that increases expected reward

**Action** A single step, in MT/LLM: a single token prediction

**State** Situation after any number of steps, in MT/LLM: the text generated so far

**Advantage** Measure of importance of a step, e.g., a game-altering move in chess

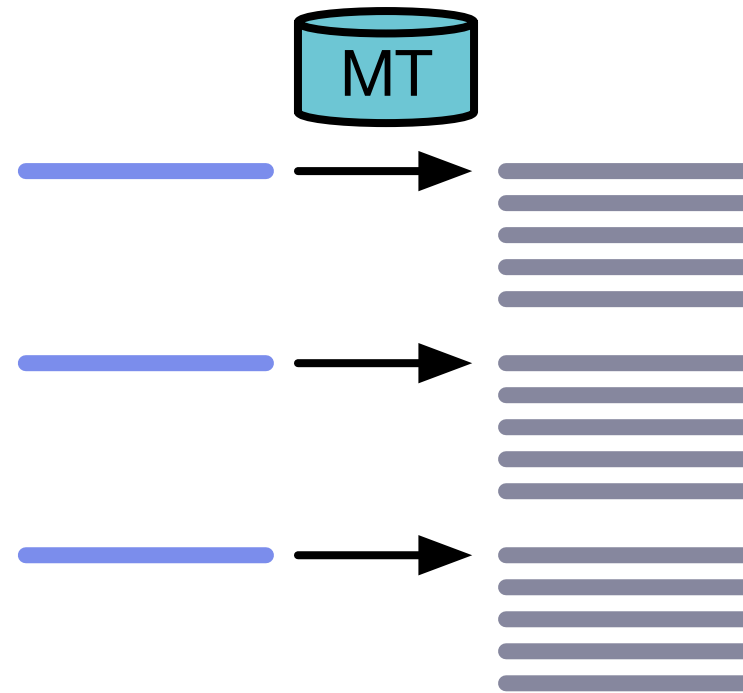
**Actor/Critic** Setup with two models: one that decides which actions to take (actor), and one that assesses outcomes (critic)

**Rollout/Trajectory** Completion of a action sequence until the end (e.g., based on current policy) — this is typically done for MT/LLM

- Reinforcement learning for MT/LLM:  
no intermediate state assessment, no partial rewards

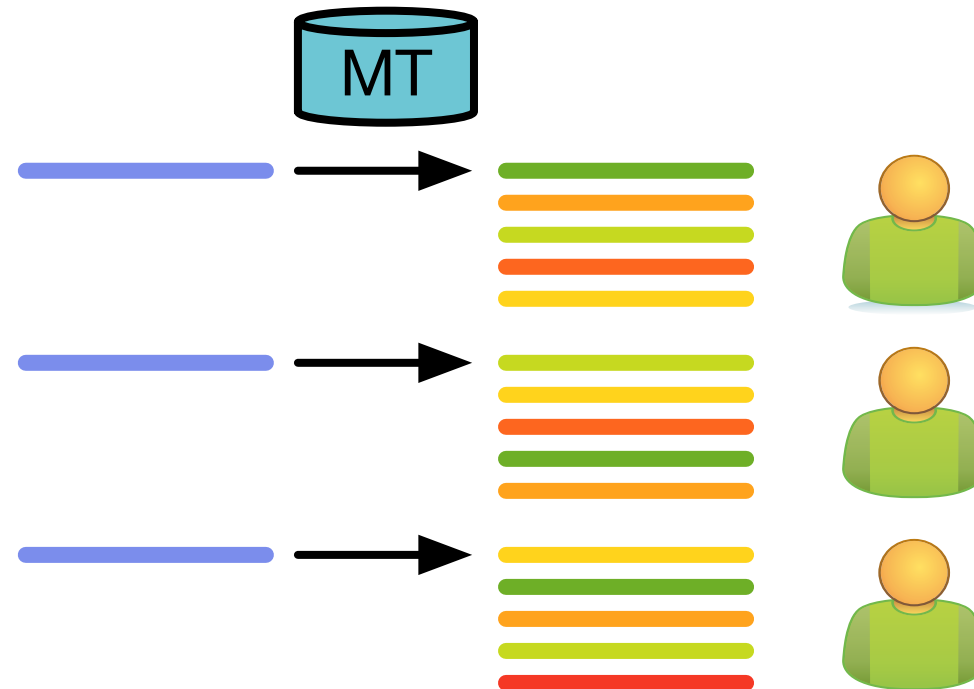
# preference training

# Learning from Human Preferences



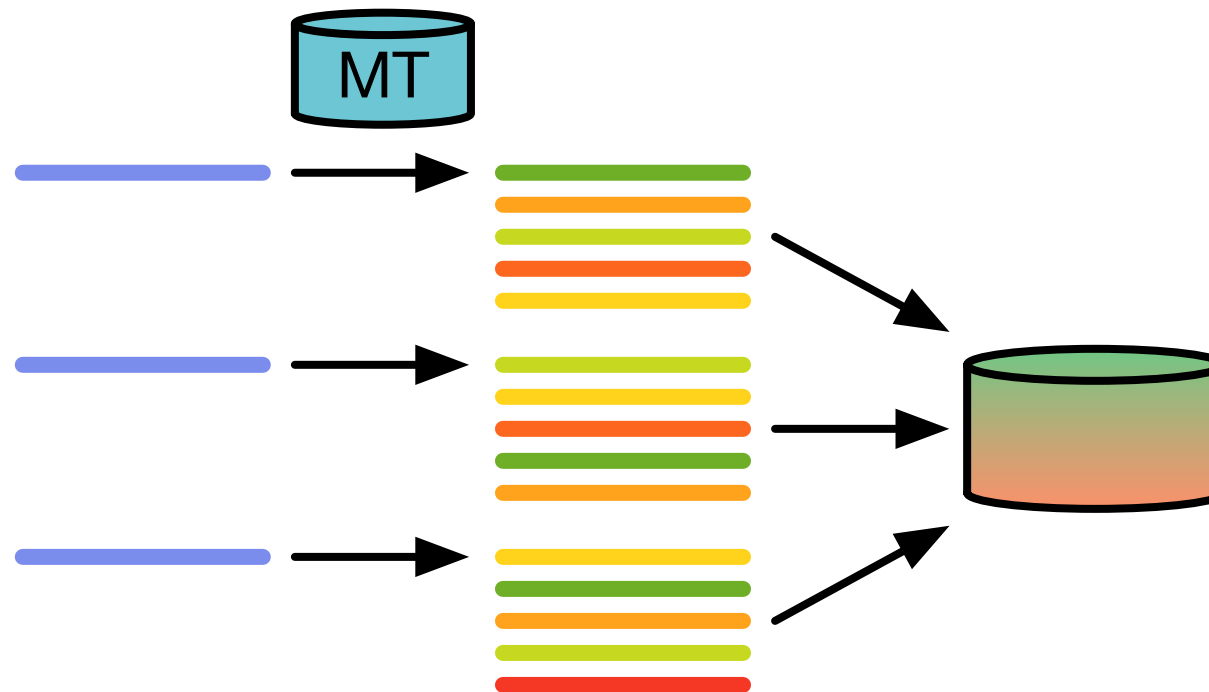
- Generate translations from a source sentence by sampling
  - greedy decoding: always choose word prediction with 80% probability
  - Monte Carlo decoding: choose it 80% of the time

# Learning from Human Preferences



- Human annotators rank the translations
- This is easier to do than authoring translations but still expensive

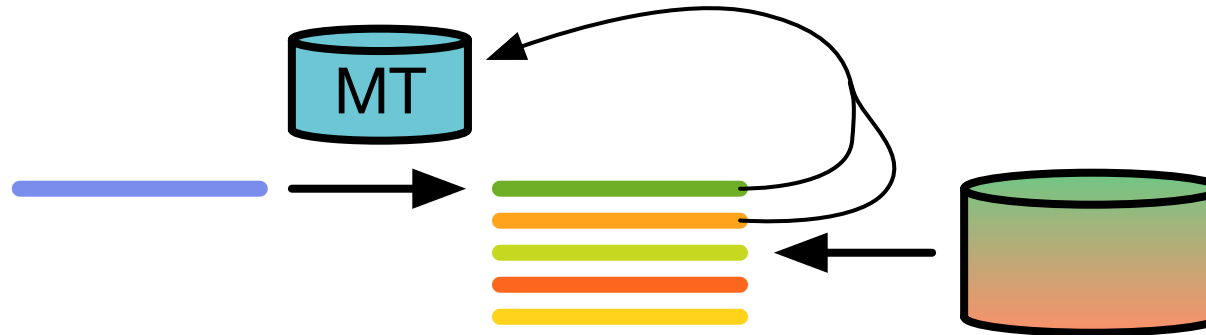
# Learning from Human Preferences



- Train a preference model
- Typically based on sequence representations from language models



# Learning from Human Preferences



- Use the preference model during training original model
  - for an input sentence, generate translations
  - score the translations with the preference model
  - update model to
    - \* promote higher-scoring translation (winner)
    - \* demote lower-scoring translation (loser)

# preference data

# Preferences for Translation

- What we need is
  - source sentence
  - multiple translations
  - human judgment which translations are better
- This seems to be expensive to obtain...

# 20 Years of WMT Evaluations

## Judge Sentence

You have already judged 14 of 3064 sentences, taking 86.4 seconds per sentence.

**Source:** les deux pays constituent plutôt un laboratoire nécessaire au fonctionnement interne de l'ue .

**Reference:** rather , the two countries form a laboratory needed for the internal working of the eu .

| Translation                                                                               | Adequacy                                                                                                                              | Fluency                                                                                                                               |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| both countries are rather a necessary laboratory the internal operation of the eu .       | <input type="radio"/> <input type="radio"/> <input type="radio"/> <input type="radio"/> <input checked="" type="radio"/><br>1 2 3 4 5 | <input type="radio"/> <input type="radio"/> <input type="radio"/> <input type="radio"/> <input checked="" type="radio"/><br>1 2 3 4 5 |
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| <b>Annotator:</b> Philipp Koehn <b>Task:</b> WMT06 French-English                         | <input type="button" value="Annotate"/>                                                                                               |                                                                                                                                       |
| Instructions                                                                              | 5= All Meaning<br>4= Most Meaning<br>3= Much Meaning<br>2= Little Meaning<br>1= None                                                  | 5= Flawless English<br>4= Good English<br>3= Non-native English<br>2= Disfluent English<br>1= Incomprehensible                        |

- Long-running WMT evaluation campaign (since 2006)
  - participants submit translations of their system
  - human evaluators score or rank these translations

→ This is the human preference data we need

- Millions of human judgments are available



# reward model

- Core part of preference training: reward model
- Setup
  - given: source sentence, candidate translation
  - output: score how good the translation is
- This is typically trained on human preference data

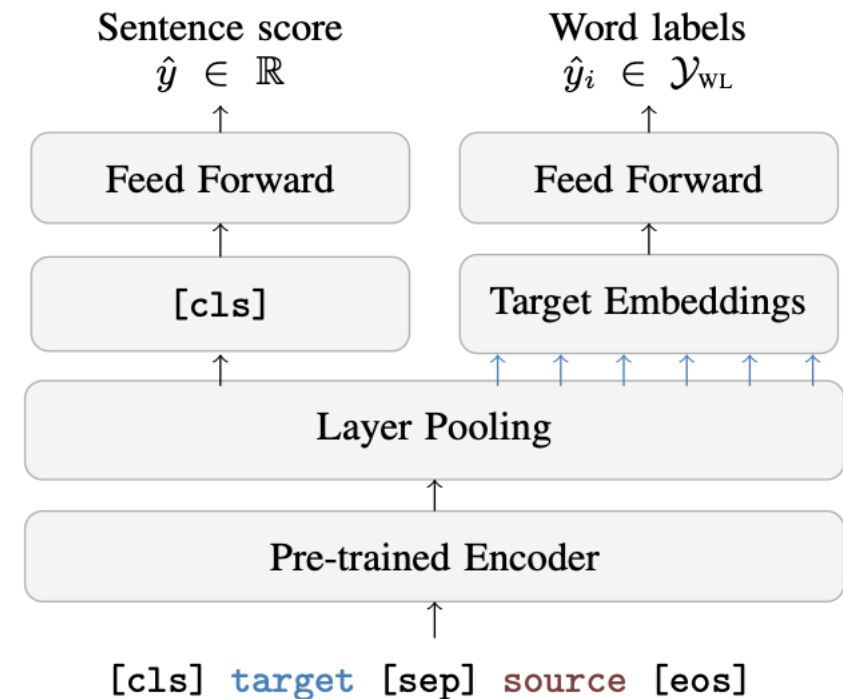
# Quality Estimation

- Long standing task in machine translation: quality estimation (or “confidence estimation”)
- For instance, for routing translations in production workflow
  - great translation → pass to customer
  - good translation → pass to professional translator
  - bad translation → ignore



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  - output: score how good the translation is
- This is exactly what we need here

- One example for a quality estimation model:  
CometKiwi [Rei et al., WMT 2023]
- Trained on word-level and sentence-level human evaluation data
  - sentence-level: direct assessment scores
  - word-level: error span annotations
- Foundation model: XLM Roberta with up to 10.7 billion parameters



# loss functions

# Reinforcement Learning from Human Feedback

- This idea was originally introduced as a form of reinforcement learning
- The idea of a reward model stems from reinforcement learning
- Originally proposed method: Proximal Policy Optimization (PPO)
- Recently, simpler methods are more common
- We will take a closer look at Direct Preference Optimization (DPO)

- First train a reward model  $r(x, y)$  for source sentences  $x$  and translations  $y$
- Sample two possible translations for an input  $x$
- Score them with the reward model
  - higher scoring translation is the winner  $y^+$
  - higher scoring translation is the loser  $y^-$
- Goal: train a new LLM  $\pi_\theta$  from an original model  $\pi_{\text{ref}}$

# Preference Model

- We can convert this into a preference model using softmax

$$P(y^+ \succ y^- \mid x) = \frac{\exp(r(x, y^+))}{\exp(r(x, y^+)) + \exp(r(x, y^-))}$$

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- With  $\sigma$  being the well-known sigmoid function

# Adding Regularization

- We do not want the model  $\pi_\theta$  to go too far afield
- In other words, keep it close to the original model  $\pi_{\text{ref}}$

⇒ We need a measure how much these models differ

- A common choice: KL divergence

# Kullback-Leibler Divergence

- Textbook definition

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- Here we deal with conditional probabilities

$$D_{\text{KL}}(P \parallel Q) = \sum_{y \in \mathcal{Y}} P(y|x) \log \frac{P(y|x)}{Q(y|x)}$$

# Kullback-Leibler Divergence

- In our case, the two distributions are  $P = \pi$  and  $Q = \pi_{\text{ref}}$ , so:

$$D_{\text{KL}}(\pi \parallel \pi_{\text{ref}}) = \sum_{y \in \mathcal{Y}} \pi(y|x) \log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)}$$

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- Another way to look at this is the expected value of the log-ratio

$$D_{\text{KL}}(\pi \parallel \pi_{\text{ref}}) = \mathbb{E}_{y \sim \pi} \log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)}$$

- Combine preference and regularization (weight  $\beta > 0$ ) to optimize model  $\pi$

$$\pi^* = \arg \max_{\pi} \mathbb{E}_x \left[ \mathbb{E}_{y \sim \pi(\cdot|x)} [r(x, y)] - \beta \text{KL}(\pi(\cdot | x) | \pi_{\text{ref}}(\cdot | x)) \right]$$

- This results in the following loss function  
(see paper for full derivation)

$$\mathcal{L}_{\text{DPO}}(x, y_w, y_l) = -\log \sigma \left( \beta \log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right)$$

[from Rafailov et al., 2023]





- Computing probabilities with both
  - the reference model  $\pi_{\text{ref}}(y|x)$  and
  - the new model  $\pi_{\theta}(y|x)$is expensive
  - twice the memory requirements
  - twice the number of computations
- Simplification: only score with new model

$$\text{Loss}(x, y_w, y_l) = \log \text{sigmoid} \left( \beta \log \pi_{\theta}(y_w|x) - \beta \log \pi_{\theta}(y_l|x) \right)$$

[from Xu et al., 2024]

# Group Relative Policy Optimization (GRPO)<sup>26</sup>



- Generate a group of translations (e.g., 64) for a sentence and score each
- Compute average of scores
- Check for each translation how its score compares against the average
  - better than average: promote
  - worse than average: demote

example:  
direct quality optimization

# Example: Direct Quality Optimization

[Uhlig et al., WMT 2025]

- Core idea
  - use existing quality estimation models as reward model: CometKiwi
  - use DPO as reinforcement learning method
- Big picture
  - sample source sentences
  - generate possible target translations
  - identify winning and losing translations
  - update model with DPO

# Create a Pool of Translations

- Given a source sentence  $x$
- Generate possible translations
  - greedy search
  - sampling  $K = 40$  additional translations

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# Create a Pool of Translations

- Given a source sentence  $x$
- Generate possible translations
  - greedy search
  - sampling  $K = 40$  additional translations
- Winning translation  $y^+$ : best according to reward model
- Losing translation  $y^-$ : randomly sampled (as long it is worse than  $y^+$ )
- Repeat this process to obtain a pool of training examples

# Overall Algorithm

- Initialize model  $\pi_\theta$  from original model  $\pi_{\text{ref}}$
- Loop for several rounds:
  - generate a pool of training examples  $P$
  - loop for several epochs:
    - \* run DPO on the pool of training examples

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**Algorithm 1:** Direct Quality Optimization

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**Parameters:** preference relation  $\succ$ , number of rounds  $n$ , epochs per round  $m$ , epoch size  $d$ , learning rate  $\alpha$ , DPO regularization  $\beta$ , number of sampled translations per source  $k$

**Input:** Source language seed dataset  $S$ , reference-free QE model  $r_{QE}$ , reference model  $\pi_{\text{ref}}$

```
 $\pi_\theta \leftarrow \pi_{\text{ref}};$ 
for round  $i = 1, 2, \dots, n$  do
   $X \leftarrow \text{sample } d \text{ sentences from } S;$ 
   $P \leftarrow \emptyset;$ 
  foreach source  $x \in X$  do
     $g \leftarrow \text{Greedy}_{\pi_\theta}(x);$ 
     $Y \leftarrow \text{sample } k \text{ translations of } x$ 
      from  $\pi_\theta;$ 
     $Y_+ \leftarrow Y \cup \{g\};$ 
     $y_w \leftarrow \text{argmax}_{y \in Y_+} r_{QE}(x, y);$ 
     $Y_l = \{y' \in Y_+ | y_w \succ_x y'\};$ 
    if  $Y_l \neq \emptyset$  then
       $y_l \leftarrow \text{sample } y \in Y_l;$ 
       $P \leftarrow P \cup \{(x, y_w, y_l)\};$ 
    end
  for epoch  $j = 1, 2, \dots, m$  do
     $\pi_\theta \leftarrow \text{DPO}(\pi_\theta, \pi_{\text{ref}}, P, \alpha, \beta);$ 
  end
end
```

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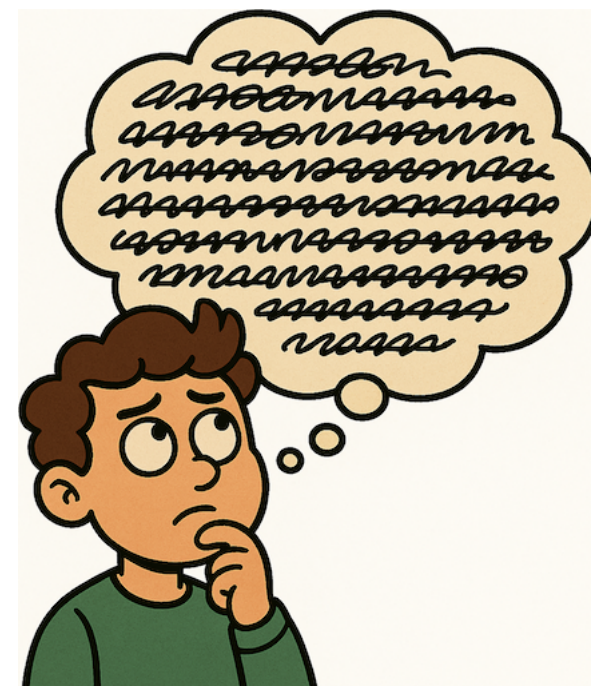
[from Ulig et al., 2025]



# reasoning models

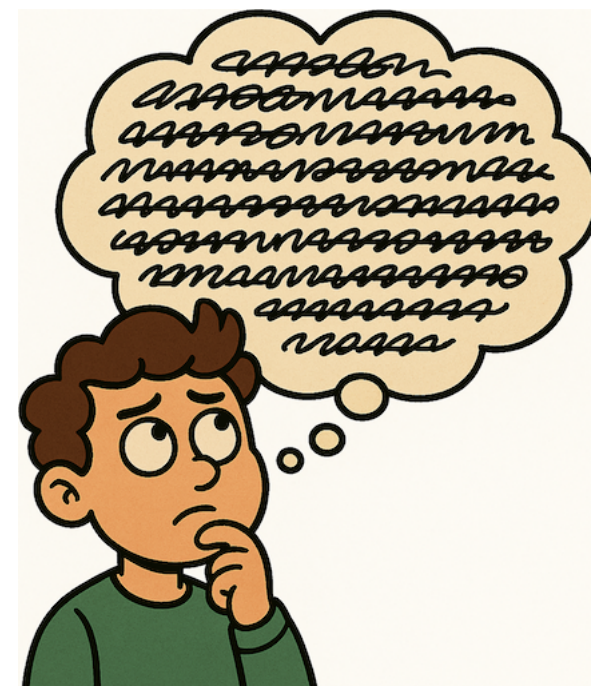
# Thinking

- Complex tasks require more “thinking”
- Typical LLM application: solving math problems
- Typical MT application: translating poetry



# Thinking

- Complex tasks require more “thinking”
- Typical LLM application: solving math problems
- Typical MT application: translating poetry
- What happens in thinking does not matter so much
- Only the final output matters



# Thinking for Translation

- Consider poetry

*"Keep, ancient lands, your storied pomp!" cries she  
With silent lips. "Give me your tired, your poor,  
Your huddled masses yearning to breathe free,  
The wretched refuse of your teeming shore.  
Send these, the homeless, tempest-tost to me,  
I lift my lamp beside the golden door!"*

- Left-to-right generation may struggle with rhyme (*she, free, me*, and *poor, shore, door*)
- Other challenges: capturing overall mood, coherent metaphors, ...
- A reasoning model is able to revise a draft



- LLM reasoning models produce output in two stages
- A thinking stage that may be hidden from the user  
*<think> Let's translate Emma Lazarus's final stanza of "The New Colossus" into German while keeping its solemn tone and rhyme scheme.  
Step 1: Analyze rhyme and rhythm Original rhyme pattern: ABABCC. A pomp,  
... </think>*
- An answer stage with just the desired output  
*<answer> „Behaltet euren Glanz und stolzen Pomp!" ruft sie  
Mit stummen Lippen. „Gebt mir eure Müden, Armen, ...  
</answer>*

# Rewards

- The reward only considers the final answer
- This may be simple binary answer checking (e.g., for math)

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- This may be simple binary answer checking (e.g., for math)
- Should we award good reasoning paths?
  - intuitively, yes
  - this is not currently done  
(except for very basic formality checks)
  - it is also very hard

questions?